MDM490

Operation Manual



MICROSENSOR

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Our company reserves the modification right for this operation manual due to renovation of production technology and craftwork. If some information is changed, no more notice will be edited.

Please pay attention to the latest version.

Our company also reserves the right of final explanation for this manual.

Version: V1.1

Thanks for your using products from MICROSENSOR. MDM490 precise piezo-resistive transmitter is one of precise instruments. We suggest you to read this manual carefully before use.

1 Introduction

MDM490 precise piezo-resistive transmitter is a differential pressure measurement device with compact size. The transmitter using a differential pressure sensor and special amplified circuit, through stability and reliability experiments, can provide standard 2-wire 4mA～20mA DC and 3-wire 0V/1V～5V DC 、0mA～10mA/20mA DC signal output by external power supply 24VDC, as well zero and span compensation in the wide temperature range. The parts being contacted with the media are the stainless steel materials and viton. The differential pressure transmitter can be used in vary industry processing control and differential pressure, flow and level measurement, etc.

2 Specifications

Pressure Range: 0mbar ~ 350mbar…35bar

Over pressure: positive pressure: ≤2 times FS

negative pressure is not allowed

Maximum Static Pressure : ≤200bar

Pressure Type: differential pressure

Accuracy: ≤±0. 5%FS

Long-term Stability:≤±0.5%FS/year（≤ 200kPa）

≤±0.2%FS/year（＞200kPa）

Power Supply: 15V~28V DC

Output: 2-wire 4mA~20mA DC

3-wire 0V/1V~5V/10V DC，0mA~10mA/20mA DC

Operation Temperature: -30℃~80℃（4-pin Plug）

-20℃~70℃（cable material: PE, PVC）

-20℃~80℃ (cable material: PUR)

Storage Temperature:-40℃~120℃，-20℃～85℃（Cable）

Response: ≤1ms

Insulation: 100MΩ@50V DC

Housing Protection: IP65

3 Outline Construction and Installation

3.1 Construction and Mounting Dimension Unit:mm

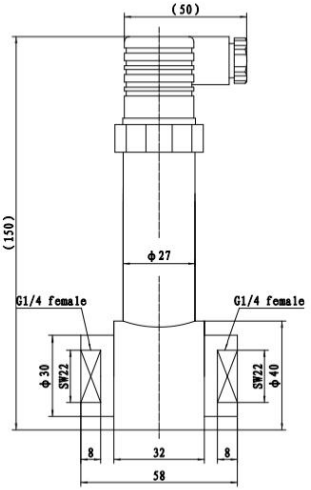


Fig.1

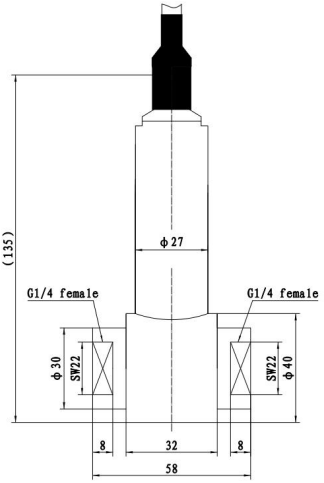


Fig.2

3.2 Installation

3.2.1 Attentions before installation:

a) The operating pressure of measuring medium is no more than the line pressure of transmitter;

b) The differential pressure of measuring medium in the worksite is no more than the maximum pressure.

c) For positive and negative pressure balance and convenient

repair, we recommend to use tri-valve （such as J23SA）to connect with the leading tube, to prevent one-side over pressure destroying the transmitter. Or the customer can buy the leading tube and tri-valve from our factory. Tri-valve operation method can see appendix “Tri-valve Operation Menu”.

3.2.2 Installation Method

a) The recommended method is to see Fig. 3.

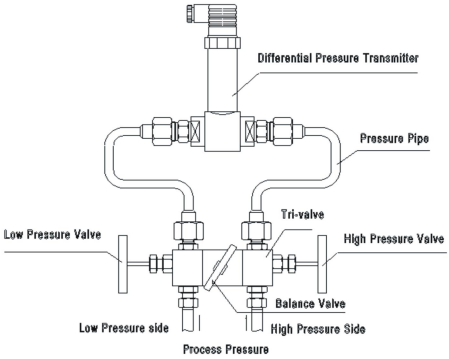


Fig. 3

b) Installation notes:

1) To prevent the installation position from affecting zero output signal, the transmitter can be installed horizontally on the tube and the plug could be installed vertically up.

2) Pay attention to whether the measured medium is consistent with the transmitter structure material.

3) Pay attention to the measuring medium is compatible with the construction material or not.Notice the positive(“＋”) and negative (“－”) mark on the pressure ports while connecting the actual pressure direction corrrectly.

4) The transmitter should be installed in the low temperature gradient and low temperature fluctuation place, to prevent vibration and shock.

5) The protection of transmitter is IP65。

6) Prevent leak, friction and temperature difference error, to prevent dregs settling on the tube affecting the diaphragm.

7) The transmitter has been calibrated when out of the factory, the customer can operate the transmitter without calibration, but checking the installation and electrical connection correct or not is necessary. The transmitter can work when excitation is connected, but the signal output is more reliable after 30 minutes.

8) Ban to pulling the cable violently, and to prevent the diaphragm damaging don’t poke the diaphragm with metal still objects.

MDM490 differential pressure transmitter allows the customers to adjust

the zero and Span output signals while you own a standard pressure controller. While adjusting, for the transmitter with plug connection, remove the socket on the top of housing (take care not to break the cable), then trim “Zero” and “Span” buttons to adjust. For the transmitter with cable connection, you need to use a wrench slightly screw the cap and you will find the Zero and Span calibrators on the top of housing. Restore them when finish.

MDM490 differential pressure transmitter has no movable mechanical parts, the housing is welded，almost no need to repair. If the transmitter is fail to work, the customer should contact with factory checking the reasons, and return to factory for repair.

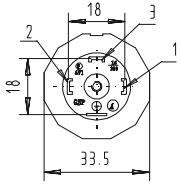
Caution：do not poke the pressure-leading hole with metal wire or something hard; and do not press the diaphragm with finger or something sharp to protect the diaphragm.

4 Electric Connection

4.1 The transmitter is connected with the outside circuit through plug or special cable.

The pin arrangements of socket are to see Fig.4, and pin definitions are as follows：

Table 1



Pin	2-wire	3-wire
1	+V	+V
2	OV/+OUT	GND
3	Null	+OUT

Fig.4

The electric definitions of cable are as follow：

Table 2

Cable	2-wire	3-wire
Red	+V	+V
Black	OV/+OUT	+OUT
White	Null	GND

4.2 Electrical connection method is indicated as follow

4.2.1 4mA~20mA DC(2-wire)

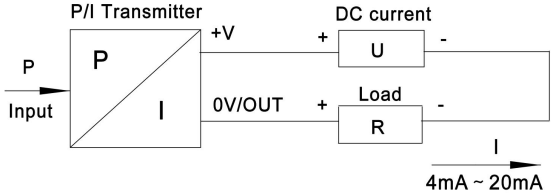


Fig. 5

4.2.2 0V/1V~5V DC(3-wire)

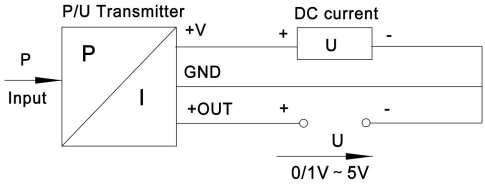


Fig. 6

4.3 Plug connection Operation Method：

a) If the plug has been connected with transmitter, loose the central bolt on the top of plug with little screwdriver and pull out the plug from transmitter.

（caution： do not take out the socket on the transmitter to protect it）

b) To disconnect the plug, please take out the central bolt on the top of plug, then turn the plug to the bottom and insert a little flat screwdriver into a corner signed “Lift” and give a force. The plug core will be disconnected with housing.

c) To connect the cable, put the cable through the cable jack and connect the wires with terminals on the plug core correctly (the connection terminals are signed with clear numbers). Please choose Φ4.5mm~7mm shield cable and connect the cable reliably to prevent short circuit.

d) Pull the cable slightly and push the plug core into housing (a rattling sound could be listened), then screw down the cable-fixed nut.

e) To remove the cable, loose the cable-fixed nut to relax the cable, and operate as Item2 to disconnect the plug. Take out the cable from terminals with little screwdriver and pull out the cable from cable-fixed nut, then renew the connection between plug core, plug and socket.

Caution: renew the rectangle-ring to the pre-mounting situation in order tomaintain the protection class when connect plug and socket.

4.4 Safe explosion proof transmitter connection Operation Method

Intrinsically safe explosion proof transmitter and safe barrier form a intrinsically safe explosion proof system, when it is operated in the explosive conditions. The safe barrier and excitation should be put in safe area, and intrinsically safe model transmitter should be put in hazardous area. Notice for connected with earth safely. To see follow Fig.7.

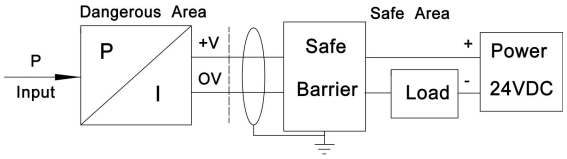


Fig. 7

≤35kPa Explosion proof grade and parameter of transmitter：

ExiaⅡBT6 Ga

Ui=28V DC Ii=93mA DC

Li≈0.1mH Ci≈0.29μF Pi=0.65W

Explosion proof grade and parameter of safety fence

ExiaⅡC

Uo=28V DC Io=93mA DC

Po=0.65W

＞35kPa Explosion proof grade and parameter of transmitter：

Ex ia Ⅱ C T6 Ga

Ui=28V DC Ii=93mA DC

Li=0mH Ci=0.012μF Pi=0.65W

Explosion proof gradeand parameter of safety fence

ExiaⅡC

Uo=28V DC Io=93mA DC

Po=0.65W

The maximum diffused capacities of cables between transmitter and safe barrier is $C_p=C_o-C_i$ ， the maximum diffused inductance is $L_C=L_o-L_i$.

The safe barrier is purchased by customers, as long as the coefficients match the requirements above and own the explosion proof certificate. Safe barrier's installation and operation should see safe barrier operation menu.

When the product is operated in "Zero"area, the excitation transformer

supplied to safe barrier should conform No. 8.1 requirements of GB/T3836.4-2021.

Changing components and construction are not permitted, maybe it will change the explosion proof specification.

5 Unpacking、Components and Storage

5.1 Unpacking

a) Be sure the package is completed, and the package should be put as the sign “UP”.

b) Be sure unpacking carefully, and prevent damaging instruments or accessories. Pay attention to the housing jacket and rubber bushing of transmitter cable.

5.2 Enclosed

The transmitter should be enclosed when out of factory：

When out-factory, the transmitter should include:

MDM490 differential transmitter	one
Electrical connection plug (insert supplied)	one
Cable (cable supplied)	1.5 meter or based on order
Product operation menu	one
Product certificate of quality	one
Quality following card	one

5.3 Storage

The transmitter should be stored in dry ventilate room, ambient

temperature -40℃~120℃,-20℃～85℃（Cable）and the relative humidity≤85%, no corrosive substance in the room.

6 Operation and Maintenance

6.1 Operation

a) The user could operate the transmitter without any adjustment. Be sure the installation and electric connection are correct.

b) The transmitter could work at once as soon as the power is supplied. But the signal output will be more stable after 30 minutes.

6.2 Maintenance

MDM490 precise piezo-resistive transmitter is the compact measurement device, please pay attention to the following items in the operation:

a) If the pressure-leading hole is jammed or the diaphragm is dirty, please clean them with impregnant which is compatible with transmitter construction material. Do not poke pressure-leading hole with hard object or brush diaphragm.

b) Other cable in the transmitter is for our company's adjustments, do not connect it at will to protect the transmitter.

c) After electric connection, make sure to screw down the socket screw-cap and cable-fixed nut.

7 Responsibility

Within one year from the delivery date, we shall repair or replace the instrument with any quality fault caused by material parts or our manufacturing technique free of charge. For non-quality malfunction during user's operation, we are in charge of repair. But the material cost and the shuttle transportation fees should be borne by users.

www.microsensorcorp.com

MICRO SENSOR CO.,LTD.

ADD: No.18 Yingda Road, Baoji, 721006, Shaanxi, P.R.China
Tel: +86(29)8834 6384 Ext.801
Fax: +86(29)8834 6384 Ext.807
E-mail: service@microsensor.cn